VIETNAM NATIONAL UNIVERSITY – HO CHI MINH CITY
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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

FINAL REPORT

Project Name: MiniCRM for Small Businesses

Course Name: Information System Management
Course ID: IT094IU_01
Lecturer: H.L.Vân
Group: 11

Group member and contributions

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5	Trương Hồng Quân	ITCSIU22275	16%	Designed the ERD and contributed to the explanation of data organization and system structure.
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1. Introduction

In small businesses, customer information is an important resource that supports communication, follow-up, and relationship management. However, customer data is often managed through simple and fragmented methods such as notebooks, spreadsheets, or separate digital records. This can make information difficult to organize, retrieve, and update consistently.

From the perspective of Information Systems Management, an effective system should support not only data storage but also the structured management of information for business use. In the context of customer management, a centralized system can improve information accessibility, support interaction tracking, and make follow-up activities more systematic.

Based on this need, this project develops a MiniCRM system for small businesses. The system is designed as a web-based prototype that allows the Admin to manage customer records, view customer details, and track interaction history in a centralized environment. Although the project is limited to core functions, it demonstrates the practical role of information systems in supporting customer information management.

This report presents the topic and business scenario of the project, the core process of the system, the main concepts involved, and the practical applications of the proposed MiniCRM system. It also discusses the system requirements, design models, implementation, testing results, and possible future improvements.

2. Topic and Business Scenario

2.1 Topic

The topic of this project is **MiniCRM for Small Businesses**. The project focuses on how an information system can support the structured storage, organization, and use of customer data in a small business environment. Instead of covering all functions of a large-scale CRM platform, the project concentrates on the essential processes of customer information management.

2.2 Business Scenario

In many small businesses, customer information is still managed through simple and fragmented methods such as notebooks, spreadsheets, or separate message records. Although these methods are easy to use at a small scale, they often become inefficient when customer data increases over time. Information may become incomplete, duplicated, or difficult to retrieve, while communication history and follow-up records may not be maintained consistently.

This situation reduces the efficiency of customer management and limits the ability of the business to maintain organized customer relationships. From an Information Systems Management perspective, this creates a clear need for a system that can centralize customer records, improve information accessibility, and support more systematic follow-up activities.

2.3 System Description

The proposed system is a web-based MiniCRM prototype designed for internal use in small businesses. The main actor of the system is the **Admin**, who manages customer records and related information. The system supports core functions such as login, dashboard access, customer management, customer detail viewing, interaction history tracking, and basic follow-up support.

The system is intended to provide a centralized environment for customer information management rather than advanced enterprise CRM functionality. Therefore, the project represents a focused academic prototype that demonstrates the practical role of information systems in supporting customer-related business processes.

3. Core Process

The core process of the proposed MiniCRM system centers on how customer information is managed in a structured and centralized manner. In the current project scope, the main actor is the **Admin**, who uses the system to access customer records, review customer-related information, and maintain interaction history. The process begins when the Admin logs into the system and accesses the dashboard, which provides an overview of customer management activities.

After entering the system, the Admin can view the customer list and select a specific customer for further review. At this stage, the system supports core customer management functions such as adding new customer records, updating existing customer information, and deleting records when necessary. This process ensures that customer data can be maintained accurately and consistently within a single system environment.

Another important part of the core process is customer detail and interaction management. Once a customer is selected, the Admin can view detailed customer information and review the interaction history associated with that customer. The Admin can also add new interaction records to support communication tracking and follow-up activities. This allows the system to function not only as a storage tool, but also as a simple support system for customer relationship management.

From an Information Systems Management perspective, this core process is significant because it organizes customer information into a clear operational flow. Instead of keeping customer data and communication records in separate places, the system integrates them into one centralized process. As a result, the Admin can manage customer-related information more efficiently, maintain continuity in customer follow-up, and support more consistent customer management practices.

Workflow

Admin Login → Access Dashboard → View Customer List → Select Customer → View Customer Detail → Review Interaction History → Add Interaction / Update Customer Information

4. Main Concepts

The proposed MiniCRM system is built on several main concepts that are closely related to customer information management in small businesses. These concepts define the role of the system as an information tool that supports data organization, communication tracking, and customer follow-up.

The first concept is **Customer Information**. This includes essential customer data such as name, contact details, status, notes, and follow-up information. In the system, such data is treated as an important business resource that must be stored and managed in a structured way.

The second concept is **Centralized Data Management**. The system is designed to keep customer-related information in one place rather than in fragmented records. This improves accessibility, reduces inconsistency, and supports more organized management activities.

The third concept is **Interaction History**. In customer management, it is important to maintain a record of previous communication and follow-up activities. Interaction history helps the Admin review past contacts and maintain continuity in customer-related actions.

The fourth concept is **Admin-based Operation**. In the current project scope, the Admin is the main actor who manages customer records and related information. This reflects the internal management role of the system and ensures that data is maintained in a controlled and consistent manner.

Overall, these concepts show that the proposed MiniCRM system is not only a simple record-keeping tool, but also a basic information system that supports customer information management in a practical business context.

5. Applications of the System

The proposed MiniCRM system can be applied in small businesses as a practical tool for managing customer-related information in a more organized and systematic way. Its first application is the **centralized storage of customer records**, which helps businesses keep important customer data in one place instead of using fragmented manual records.

A second application is **interaction tracking**. By maintaining customer interaction history, the system supports more consistent communication and follow-up activities. This is useful for small businesses that need to review previous contacts and maintain continuity in customer relationships.

The system can also support **daily customer management operations**. Through functions such as viewing customer details, updating records, and adding interaction information, the system helps the Admin manage customer data more efficiently and reduce the difficulty of retrieving information when needed.

From a broader Information Systems Management perspective, the system demonstrates how a simple information system can improve information accessibility, support operational control, and contribute to more effective customer management practices in small businesses.

6. Functional Requirements

The functional requirements of the proposed MiniCRM system describe the main functions that the system must provide to support customer information management

in a small business context. In the current project scope, these requirements are centered on the activities performed by the **Admin** as the main actor of the system.

FR1. Login
The system shall allow the Admin to log into the system using registered account credentials.

FR2. Dashboard Access
The system shall allow the Admin to access the dashboard after successful login.

FR3. View Customer List
The system shall allow the Admin to view the list of stored customer records.

FR4. Add Customer
The system shall allow the Admin to add a new customer record into the system.

FR5. Edit Customer Information
The system shall allow the Admin to update existing customer information.

FR6. Delete Customer
The system shall allow the Admin to delete a customer record from the system.

FR7. View Customer Detail
The system shall allow the Admin to view detailed information for a selected customer.

FR8. View Interaction History
The system shall allow the Admin to review the interaction history associated with a selected customer.

FR9. Add Interaction
The system shall allow the Admin to add a new interaction record for a selected customer.

These functional requirements represent the core operational functions of the current MiniCRM prototype. Together, they support the main process of storing, accessing, updating, and tracking customer-related information in a centralized environment.

7. Non-functional Requirements

The non-functional requirements of the proposed MiniCRM system describe the quality attributes that support its practical use in a small business environment. Although the system is developed as an academic prototype, these requirements are important in ensuring that the system is usable, consistent, and appropriate for customer information management.

NFR1. Usability
The system should provide a simple and user-friendly interface for the Admin.

NFR2. Browser Compatibility
The system should operate properly on common web browsers in a standard desktop environment.

NFR3. Performance
The system should provide reasonable response time during normal user operations.

NFR4. Consistency
The system should display customer-related information in a clear and consistent format across different pages.

NFR5. Maintainability
The system should be structured in a way that allows future modification and extension with reasonable effort.

NFR6. Data Organization
The system should maintain customer and interaction data in a structured and organized manner.

NFR7. Basic Security
The system should require login before allowing access to the main system functions.

These non-functional requirements support the role of the MiniCRM system as a simple but effective information management tool for small businesses.

8. Use Case Diagram

The Use Case Diagram of the proposed MiniCRM system illustrates the main functions performed by the **Admin**, who is the primary actor in the current project scope. The diagram is used to represent the interaction between the Admin and the core functions of the system in a clear and simplified way.

In this project, the Use Case Diagram includes the following main use cases: **Login**, **View Dashboard**, **View Customers**, **Add Customer**, **Edit Customer**, **Delete Customer**, **View Customer Detail**, **View Interaction History**, and **Add Interaction**. These use cases reflect the essential activities required for customer information management in the proposed MiniCRM system.

From an Information Systems Management perspective, the Use Case Diagram is important because it shows how the system supports operational tasks related to customer data organization, customer record maintenance, and interaction tracking. It also helps clarify the scope of the system by focusing on the core functions that are relevant to small business customer management.

In the current version of the project, the diagram is intentionally limited to one main actor, the Admin, in order to match the scope of the system and maintain a clear focus on internal customer information management activities.

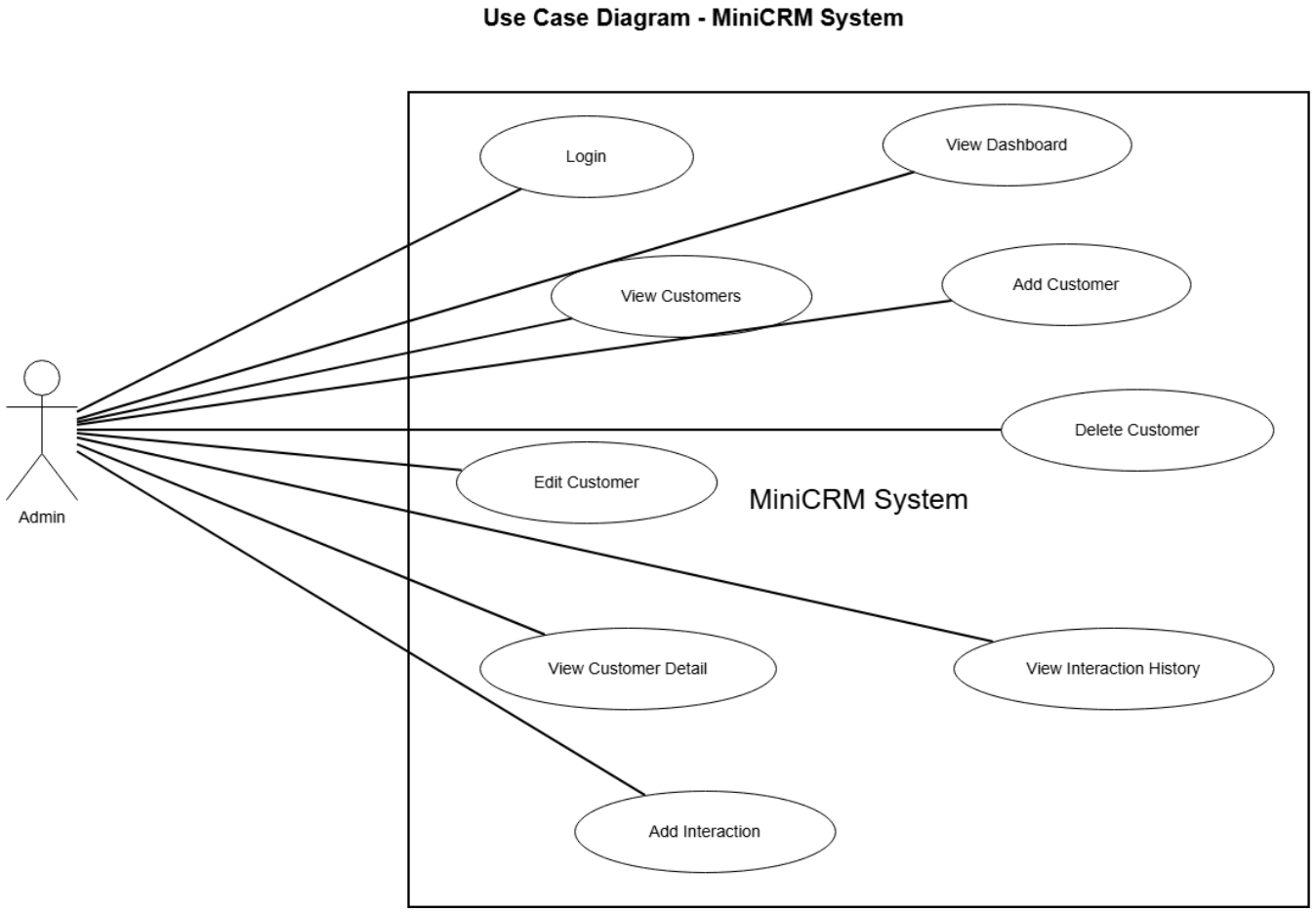

Figure 1. Use Case Diagram of the proposed MiniCRM system

9. ERD / Database Design

The ERD of the proposed MiniCRM system represents the main data structure used to support customer information management. In the current project scope, the database design focuses on organizing customer-related data in a structured and centralized manner so that the system can support customer record management, interaction tracking, and basic follow-up activities.

The design includes four main entities: **Admin**, **Customers**, **Interactions**, and **Tasks**. Among these entities, **Customers** is the central entity because the main purpose of the system is to store and manage customer information. The **Admin** entity represents the internal actor who manages customer-related records. The **Interactions** entity is used to record communication history and follow-up activities associated with each customer. The **Tasks** entity is included as a supporting component for follow-up management, although it is not the main focus of the current system scope.

The relationships among these entities reflect the operational logic of the system. One Admin can manage many customer records. Each customer can have multiple interaction records, which allows the system to maintain communication history over time. A customer may also be associated with multiple task records for supporting follow-up purposes. This structure helps organize customer data in a consistent and accessible way.

From an Information Systems Management perspective, the ERD is important because it shows how business information is structured to support system functions. Instead of storing customer information in fragmented records, the proposed database design provides a more systematic way to manage customer profiles, interaction history, and related operational data.

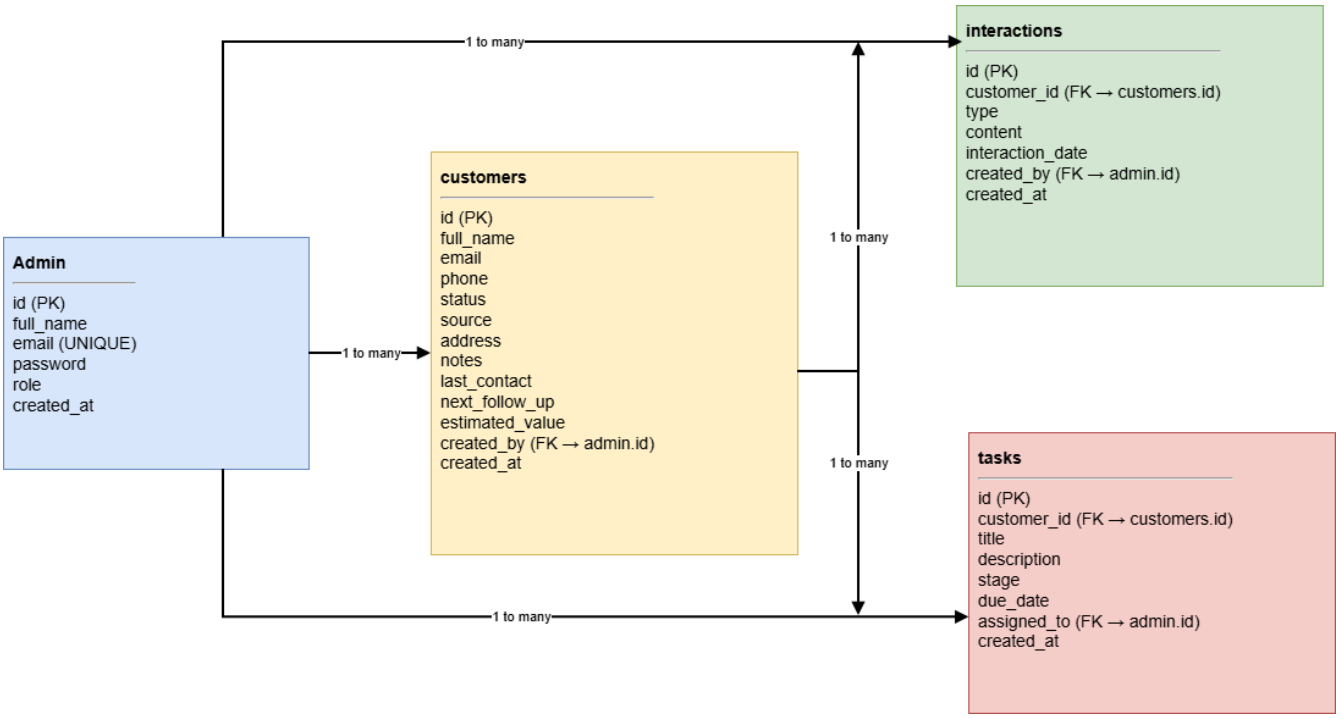

Figure 2. ERD of the proposed MiniCRM system

10. System Architecture / Technology Stack

The proposed MiniCRM system is developed as a **web-based information system** for customer information management in small businesses. In the current project, the system follows a basic multi-layer structure consisting of the **presentation layer**, **application layer**, and **data layer**. This structure allows the system to support customer-related operations in a more organized and systematic manner.

The **presentation layer** is implemented through the user interface pages, including the login page, dashboard, customer list page, customer form page, and customer detail page. This layer allows the Admin to access the system functions and interact with customer-related information.

The **application layer** is implemented using **Node.js** and **Express.js**. This layer is responsible for handling system logic, processing requests, supporting customer and interaction operations, and connecting the interface with the data storage layer.

The **data layer** is implemented using **MySQL**, which is used to store structured data related to customers, interactions, and supporting records. Through this layer, the system is able to organize customer information in a more centralized and consistent way.

The main technologies used in the project are **HTML, CSS, JavaScript, Node.js, Express.js, and MySQL**. Together, these technologies provide an appropriate foundation for implementing the proposed MiniCRM system and demonstrating how an information system can support customer information management in a small business environment.

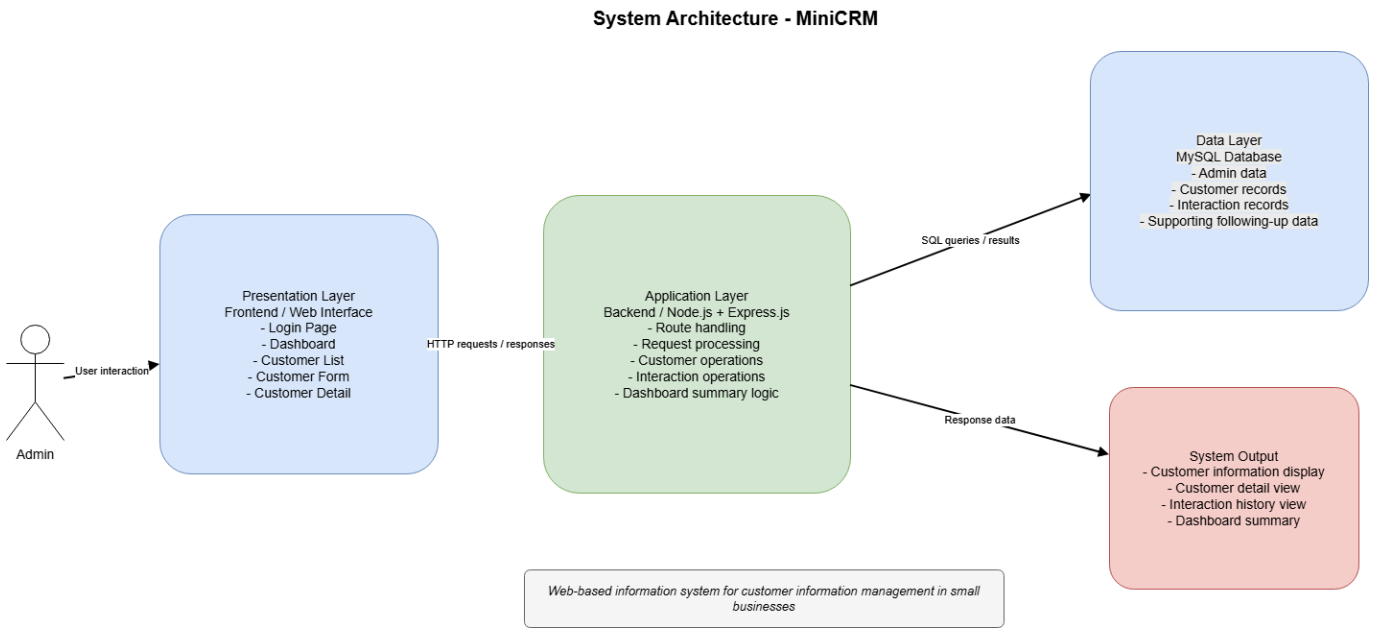

Figure 3. System Architecture of the proposed MiniCRM system

11. Implementation

The implementation of the proposed MiniCRM system focuses on the development of the main interface pages and customer management functions defined in the project scope. In the current version, the system is implemented as a web-based prototype that demonstrates the core process of customer information management for small businesses.

The implemented system includes several main pages, namely the **login page**, **dashboard**, **customer list page**, **customer form page**, and **customer detail page**. These pages are designed to support the main operational flow of the system, from system access to customer record management and interaction tracking.

In functional terms, the implementation supports the core activities of the **Admin**, including logging into the system, viewing customer records, adding new customers, editing customer information, deleting customer records, viewing customer details, reviewing interaction history, and adding interaction records. Through these functions, the system demonstrates how customer-related information can be managed in a more structured and centralized way.

From an Information Systems Management perspective, the implementation is important because it translates the proposed analysis and system design into a working prototype. It shows how the identified requirements, use cases, and data structure can be represented in an actual system interface and operational flow.

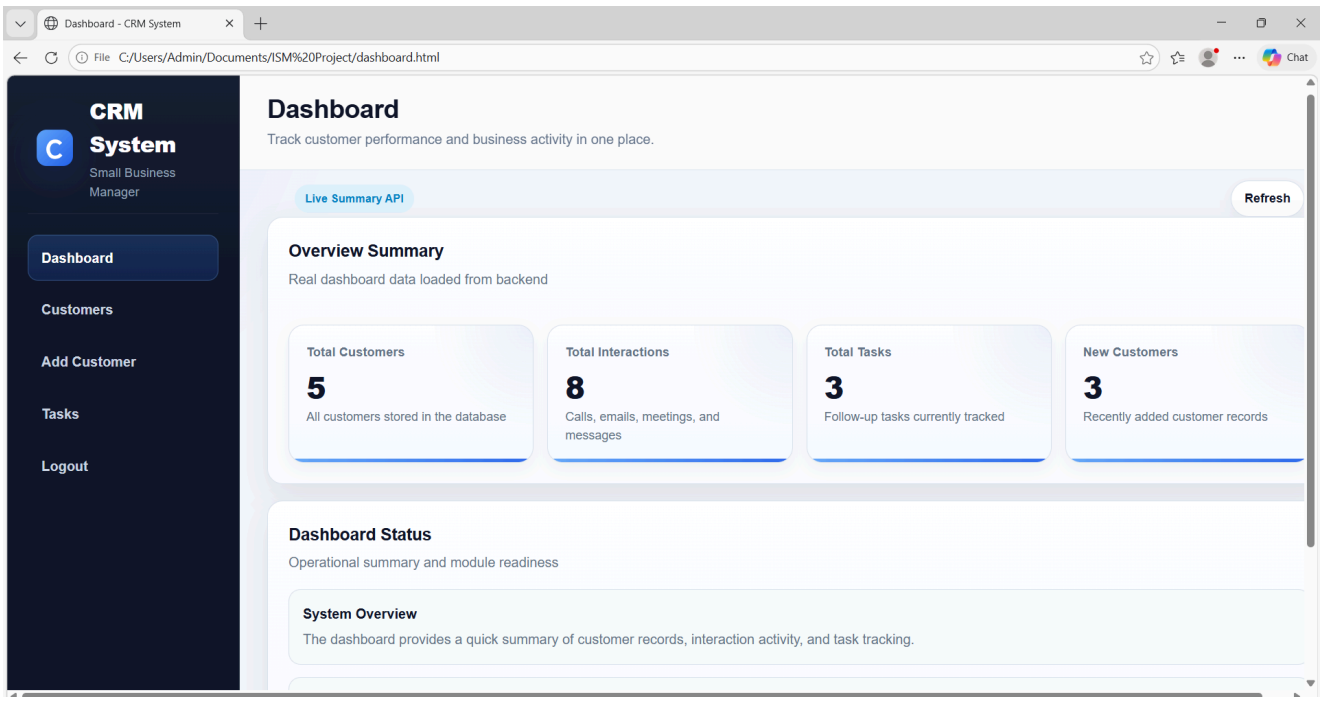

Figure 4. Dashboard interface of the proposed MiniCRM system

12. Testing and Results

Testing was conducted to verify whether the proposed MiniCRM system performs its core functions correctly according to the defined project scope. The testing process focused on the main activities of the Admin, including login, dashboard access, customer list viewing, customer creation, customer update, customer deletion, customer detail viewing, interaction history viewing, and interaction creation.

The testing results indicate that the main system functions operate as expected. The Admin is able to access the dashboard after login, retrieve and manage customer records, view detailed customer information, and maintain interaction history through the implemented interface pages. In addition, the system is able to support the addition of new customer records and new interaction records in a consistent operational flow.

From an Information Systems Management perspective, these results are significant because they show that the proposed MiniCRM system is not only conceptually designed, but also functionally demonstrated in practice. The implemented prototype confirms that the identified requirements, use cases, and data structure can be translated into a workable system for customer information management in a small business context.

The overall testing results confirm that the proposed MiniCRM prototype is able to support its intended customer information management process in a structured and usable manner.

13. Limitations and Future Improvements

The proposed MiniCRM system is developed as an academic prototype with a focused scope on core customer information management functions for small businesses. In its current version, the system emphasizes the essential process of managing customer records, viewing customer details, and tracking interaction history in a centralized environment. This focused scope is appropriate for demonstrating the practical role of an information system in supporting customer-related operations.

Future development may extend the system through broader integration, more advanced data processing capabilities, and additional support functions for customer follow-up and monitoring. Such improvements would further enhance the system’s practical value in a real business setting.

Overall, the current version provides a clear and workable foundation for customer information management, while future development can expand its functional depth and system capability.

14. Conclusion

In conclusion, the proposed MiniCRM system demonstrates how a simple information system can support customer information management in a small business context. By focusing on core functions such as customer record management, customer detail viewing, and interaction history tracking, the system provides a structured and centralized approach to handling customer-related information.

From an Information Systems Management perspective, the project highlights the importance of organizing business information in a way that supports operational activities and improves information accessibility. Although the system is developed as an academic prototype, it still reflects the practical value of information systems in supporting more systematic customer management.

Overall, the project achieves its main objective of proposing and demonstrating a MiniCRM solution for small businesses. It shows that even a focused and simple system can contribute meaningfully to customer information organization and customer-related process support.

15. Appendix

This appendix provides supplementary materials that support the analysis, design, implementation, and testing of the proposed MiniCRM system. It includes relevant diagrams, interface screenshots, and other supporting project evidence.

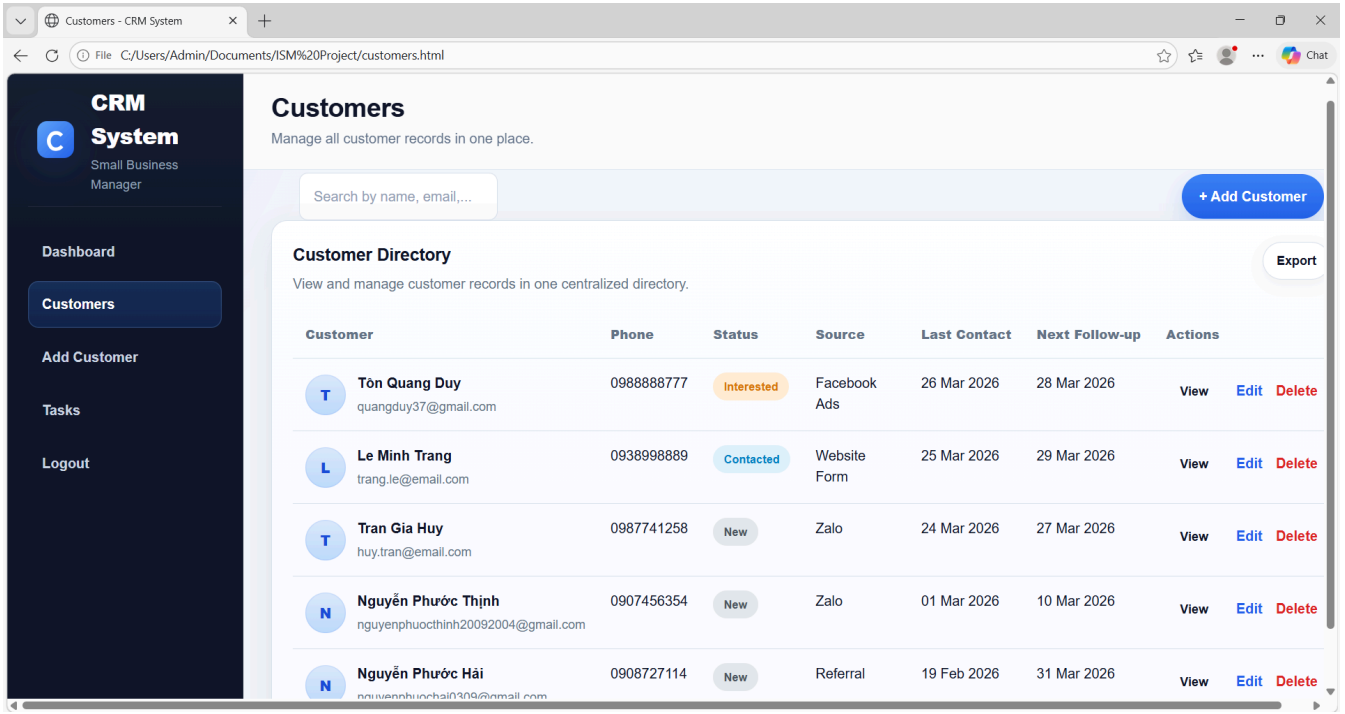

Figure 5. Customer List interface of the proposed MiniCRM system

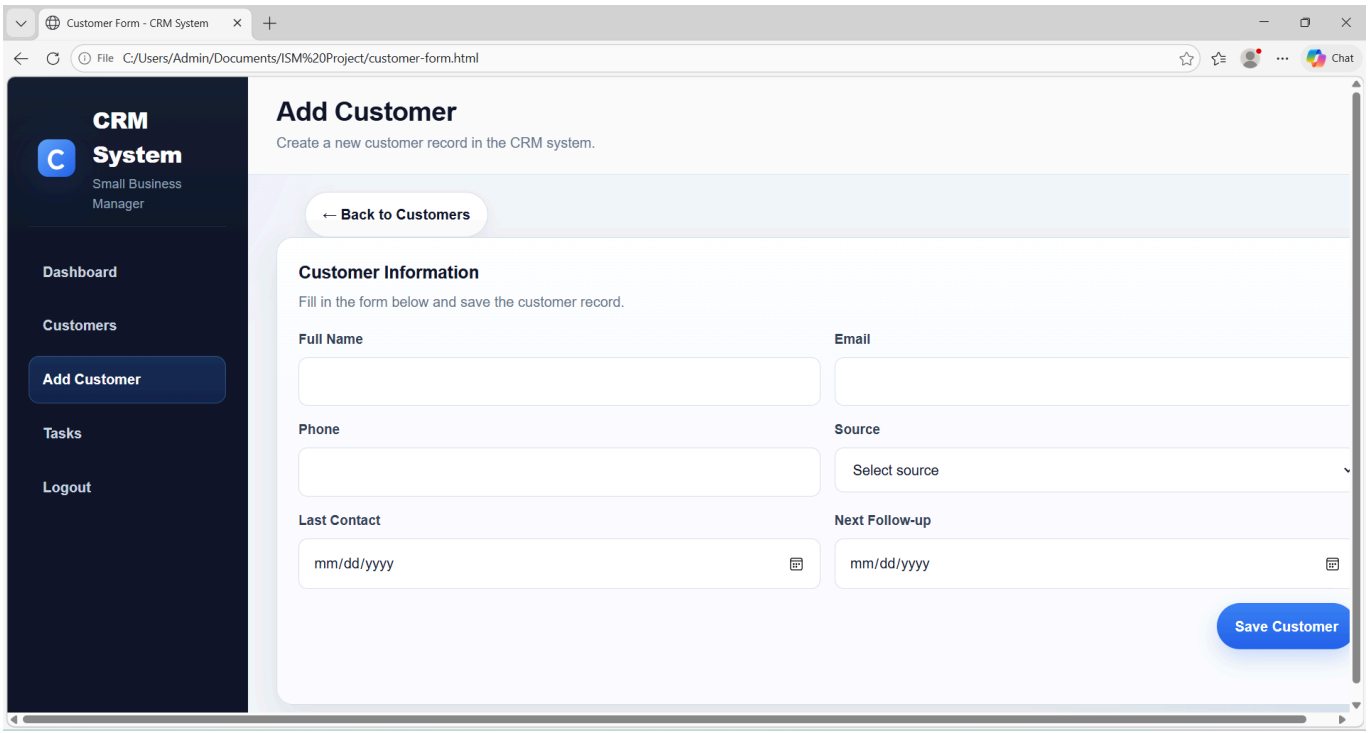

Figure 6. Customer Form interface of the proposed MiniCRM system

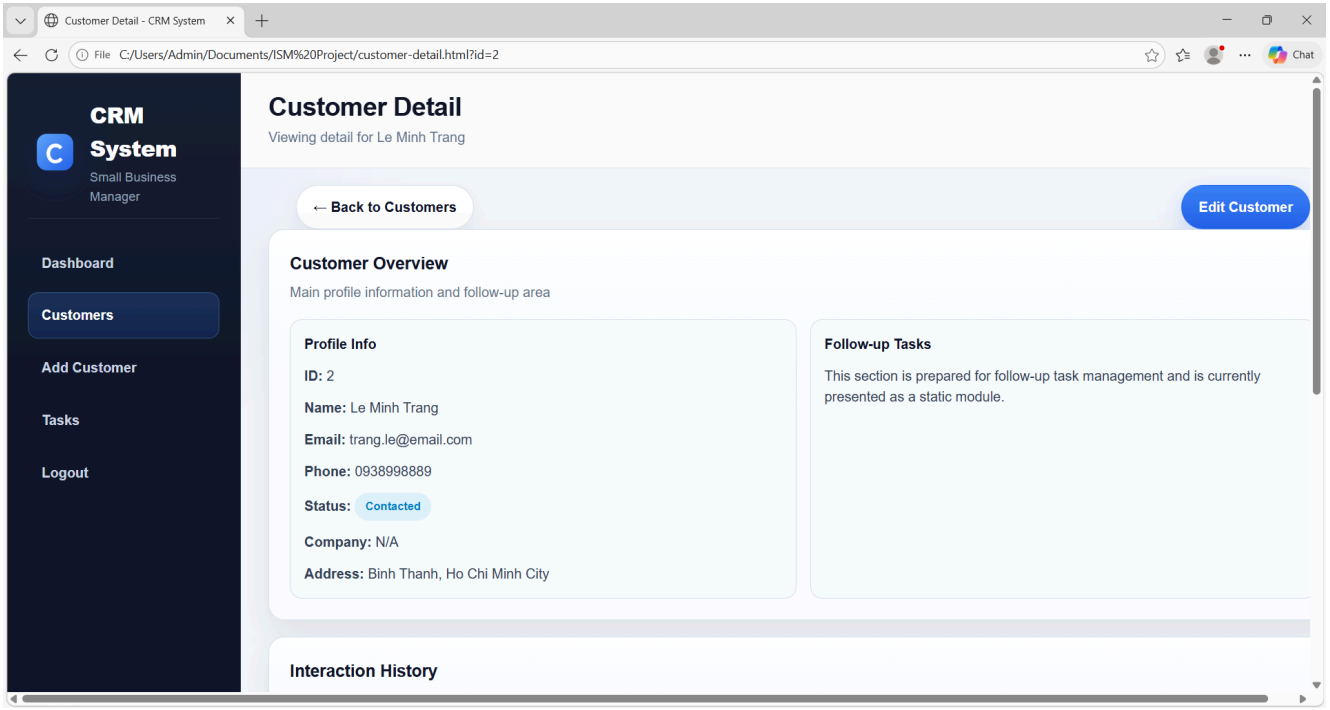

Figure 7. Customer Detail interface of the proposed MiniCRM system

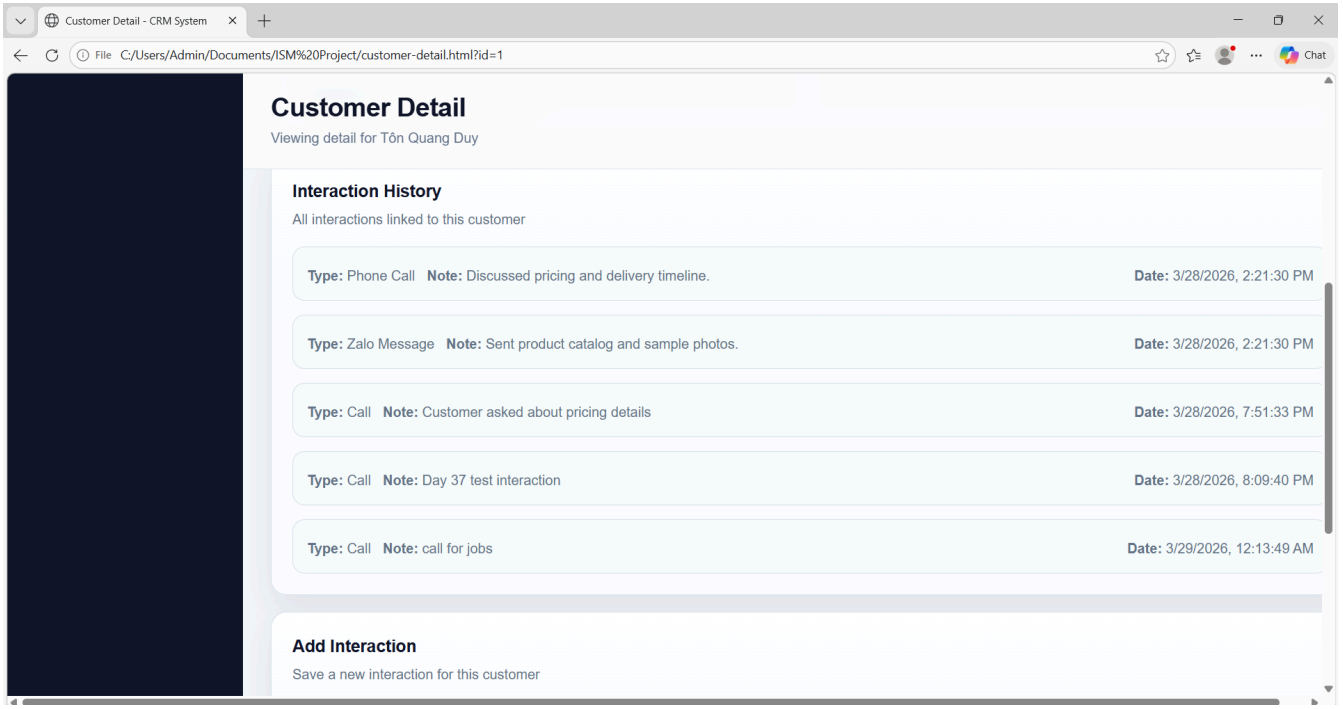

Figure 8. Interaction History section of the proposed MiniCRM system